

Clinical Trials as Multi-Expert Audit: Phase III, Meta-Analysis, and the Cercis Foundation of Evidence-Based Medicine

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Abstract

Evidence-based medicine (EBM) rests on a hierarchy of evidence: systematic reviews at the apex, followed by randomized controlled trials (RCTs), cohort studies, case series, and finally expert opinion at the base. We prove that this hierarchy is precisely a spectrum of **audit depth** in the SCX framework, with clinical trials functioning as principled multi-expert ($M > 1$) audits of treatment efficacy. The core contributions are nine formal results establishing clinical trials as the original and most consequential application of multi-expert audit theory. (1) A Phase III randomized controlled trial is an **M -expert audit** where patients, investigators, and data monitors collectively assess the treatment effect, with blinding and randomization serving as gauge-fixing mechanisms. (2) The placebo control arm is **gauge zero**: placebo response is a gauge artifact that arises from patient and observer expectations, and blinding removes it by making the treatment assignment gauge-invariant. (3) Meta-analysis computes the **Cercis score** across independent trials: the overall effect estimate is the quality-weighted average, and the heterogeneity statistic I^2 measures **gauge misalignment** across studies. (4) Diagnostic disagreement among radiologists is a **high-Cercis signal**: second opinions correspond to an $M + 1$ audit that exponentially reduces the false-positive rate. (5) The EBM hierarchy is an **audit depth function** $\text{AuditDepth}(\text{study type})$ that maps study design to the effective number of independent expert auditors. (6) Eminence-based medicine (“because I said so”) is the $M = 1$ catastrophe: a single expert self-certifies without external audit, analogous to a gatekeeper with infinite concealment. (7) The reproducibility crisis is an **audit failure**: p -hacking corresponds to gauge manipulation, and publication bias corresponds to censored audit data. (8) p -hacking shifts the audit gauge, inflating the false-positive rate by exactly the factor predicted by the SCX gauge transformation theory. (9) Cercis-based trial design gives the optimal number of experts M^* for a given effect size and desired statistical power. A complete numerical verification suite is provided in `verify_medicine_audit.py`.

1 Introduction

Clinical medicine confronts a fundamental epistemic challenge: how does one know whether a treatment works? The answer, developed over the past seven decades, is the randomized controlled trial [?, ?]. An RCT enrolls patients, randomly assigns them to treatment

or control, blinds participants and investigators to the assignment, and measures a pre-specified outcome. The statistical test at the end—did the treatment group do better than chance?—is the definitive arbiter.

This is an audit problem.

The SCX framework (State-Conditioned eXpertise) [?] provides a formal theory of multi-expert consensus, bias detection, and information concealment. Its core insight is that the Cercis score $S(s) = Q(s) + \eta(t) \cdot N(s)$ decomposes any audited state into a quality (agreement) component and a novelty (distinguishability) component. An expert with a non-zero gatekeeper parameter conceals their true state behind an information barrier. In this paper, we show that the entire edifice of evidence-based medicine—from the Phase III RCT to systematic review and meta-analysis—is a manifestation of the SCX audit framework, with clinical trials serving as the paradigmatic $M > 1$ audit.

The central claim is that the evidence hierarchy of EBM is not merely a convenient ranking of study designs but a mathematically precise **audit depth hierarchy**. At the top, systematic reviews and meta-analyses aggregate information across many independent trials ($M \gg 1$). At the bottom, expert opinion is a single expert’s assertion ($M = 1$), which the SCX framework identifies as fundamentally unaudited—it is eminence-based medicine, not evidence-based medicine.

Why this reframing matters. The mapping from clinical trials to SCX audit yields quantitative insights:

leftmargin=*, label=(1) **Placebo as gauge artifact.** The placebo response is not a nuisance to be minimized but a **gauge artifact**—an expectation bias that the blinding procedure (our gauge-fixing transformation) removes. This explains why placebo effects are largest in subjective outcomes and vanish under objective measurements: subjective outcomes are gauge-dependent.

leftmargin=*, label=(2) **Meta-analysis as Cercis computation.** The standard meta-analytic fixed-effect estimate $\hat{\theta}_{\text{FE}}$ is exactly the quality-weighted Cercis score across trials, with weights inversely proportional to the variance (inverse-variance weighting). Random-effects meta-analysis corresponds to the Cercis annealing schedule $\eta(t)$.

leftmargin=*, label=(3) **I^2 as gauge misalignment.** The heterogeneity statistic I^2 —the proportion of variance due to between-study differences rather than chance—is exactly the **gauge misalignment measure** across the expert ensemble.

leftmargin=*, label=(4) **Second opinion as $M + 1$ audit.** Diagnostic disagreement among radiologists ($\kappa = 0.4$ – 0.6 in mammography screening) is not a failure of medicine but a **high-Cercis signal**: the disagreement rate equals the rate at which a second expert detects gauge-invariant information that the first expert missed.

Structure of the paper. Section ?? establishes Phase III trials as $M > 1$ audits. Section ?? proves the placebo-as-gauge-zero theorem. Section ?? formalizes meta-analysis as Cercis computation. Section ?? analyzes diagnostic disagreement as the Cercis signal. Section ?? maps the EBM evidence hierarchy to audit depth. Section ?? proves the $M = 1$ catastrophe of eminence-based medicine. Section ?? diagnoses the reproducibility crisis as audit failure. Section ?? gives Cercis-based trial design. Section ?? discusses implications and open problems.

2 Phase III Trials as $M > 1$ Audit

2.1 The Architecture of the Randomized Controlled Trial

Definition 2.1 (Clinical Trial as Audit Configuration). A *clinical trial audit configuration* is a tuple $\mathcal{T} = (\mathcal{P}, \mathcal{I}, \mathcal{D}, \mathcal{E})$ where:

leftmargin= $\mathcal{P} = \{p_1, \dots, p_n\}$ is the set of **patients**, each providing an outcome measurement $y_i \in \mathbb{R}$;*

leftmargin= $\mathcal{I} = \{i_1, \dots, i_m\}$ is the set of **investigators**, each providing assessments, measurements, and interpretations of the outcomes;*

leftmargin= $\mathcal{D} = \{d_1, \dots, d_k\}$ is the **data monitoring** infrastructure (DSMB, statisticians, auditors) that checks integrity;*

leftmargin= $\mathcal{E}$ is the **evidence function** mapping the trial data to a summary statistic $\hat{\theta}$ (e.g., log odds ratio, hazard ratio, mean difference) with standard error σ .*

The **expert ensemble** of the trial is the union $\mathcal{E}_{\text{trial}} = \mathcal{P} \cup \mathcal{I} \cup \mathcal{D}$, and the **audit multiplicity** is $M = |\mathcal{E}_{\text{trial}}|$.

The key insight is that a clinical trial is not a single expert's opinion but an assembly of many interdependent experts. Each patient is an expert on their own symptoms. Each investigator is an expert on measurement. Each data monitor is an expert on integrity. The statistical test aggregates these expert assessments into a single decision: does the treatment work?

Theorem 2.2 (RCT as $M > 1$ Audit). *[Full Proof]* A properly conducted Phase III RCT with n patients, m investigators, and k data monitors is an M -expert audit with:

$$M = n_{\text{eff}} + m_{\text{eff}} + k_{\text{eff}}, \quad (1)$$

where n_{eff} is the effective number of independent patient outcome reports (after accounting for outcome correlation), m_{eff} is the effective number of independent investigator assessments, and k_{eff} is the effective number of independent monitoring decisions. The **Cercis** score of the trial is:

$$S(\mathcal{T}) = -\log \mathbb{P}(\hat{\theta} \geq \hat{\theta}_{\text{obs}} \mid H_0 : \theta = 0), \quad (2)$$

which is exactly the Shannon information (surprisal) of the observed treatment effect under the null hypothesis of no effect.

Proof. Step 1: Each patient as an expert. Each patient p_i in the treatment arm reports an outcome $y_i^{(t)}$, and each patient in the control arm reports $y_i^{(c)}$. Under the null hypothesis H_0 , the expected difference is zero. The aggregate test statistic $\hat{\theta} = \bar{y}^{(t)} - \bar{y}^{(c)}$ is a one-dimensional summary of n independent expert reports. The Cercis quality score is the negative log-probability of observing $\hat{\theta}$ under H_0 :

$$Q = -\log \mathbb{P}(|\hat{\theta}| \geq |\hat{\theta}_{\text{obs}}| \mid H_0). \quad (3)$$

Step 2: Randomization as gauge fixing. Without randomization, the treatment assignment $Z_i \in \{0, 1\}$ is confounded with patient characteristics X_i . In SCX language, the confounding transforms the gauge: we cannot distinguish the treatment effect from

the selection effect. Randomization ensures $Z_i \perp X_i$, which is the **gauge-fixing condition**: the treatment assignment is uniform across all observed and unobserved patient characteristics.

Step 3: Blinding as information control. Patient blinding ensures that the patient-expert’s outcome report is not influenced by knowledge of treatment assignment. Investigator blinding ensures the same for the investigator-expert. Double-blinding is the condition that both patient-expert and investigator-expert gauge parameters are zero:

$$g_P = g_I = 0. \quad (4)$$

This is the ideal audit condition: all experts report their outcomes without bias from knowledge of the treatment assignment.

Step 4: Audit multiplicity. The effective audit multiplicity is lower than the raw number of participants because within-group outcomes are correlated (patients in the same center, under the same investigator). The correction factor is the design effect $\text{DEFF} = 1 + \rho(\bar{n} - 1)$, where ρ is the intra-cluster correlation and $\bar{n}$ is the mean cluster size. The effective number of independent expert reports is:

$$n_{\text{eff}} = \frac{n}{\text{DEFF}}. \quad (5)$$

The audit multiplicity M is then the sum of all effective independent experts in the trial. $\square$

Corollary 2.3 (Intention-to-Treat as Audit Completeness). *Intention-to-treat analysis—analyzing patients according to their randomized assignment regardless of actual treatment received—is the **audit completeness principle**: all randomized experts must be included in the audit, regardless of protocol deviations. Excluding non-compliant patients is equivalent to audit censoring, which biases the *Cercis* score by selectively removing disagreeing expert reports.*

2.2 Randomization and the Gatekeeper Parameter

The gatekeeper parameter g in the SCX framework measures the concealable bias of an expert. In clinical trials, the analogous quantity is the **confounding bias**: the difference between the naive treatment effect estimate and the true causal effect.

Proposition 2.4 (Randomization Zeroes the Gatekeeper). *Under perfect randomization, the gatekeeper parameter of the trial configuration is zero: $g_T = 0$. Under observational study designs (cohorts, case-control), $g_T > 0$ and grows with the number and severity of uncontrolled confounders. The **residual gatekeeper norm** after randomization is bounded by:*

$$\|g_T\| \leq \frac{\|\Gamma\|}{\sqrt{n}}, \quad (6)$$

where Γ is the vector of covariate imbalances (standardized mean differences) across treatment arms, and n is the sample size.

3 Placebo Control as Gauge Zero

3.1 The Placebo Response

The placebo response—improvement in patients receiving an inert intervention—is one of the most robust phenomena in clinical medicine [?, ?]. Placebo analgesia alone produces effect sizes of $d = 0.5$ – 0.8 in pain trials, comparable to many active analgesics.

In the SCX framework, the placebo response is not a biological fact but a **gauge artifact**. It arises because the patient-expert knows they are in a therapeutic context, and this knowledge shifts their outcome reporting. The true treatment effect—the gauge-invariant quantity—is the difference between the treatment group outcome and the placebo group outcome.

Definition 3.1 (Placebo Response as Gauge Transformation). *Let $g_{\text{expectation}}$ be the gatekeeper parameter encoding the patient’s expectation of benefit. The observed outcome in any treatment arm is:*

$$y_{\text{obs}} = y_{\text{true}} + \varphi(g_{\text{expectation}}), \quad (7)$$

where $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ is the **gauge transformation function** mapping the expectation gatekeeper to the outcome shift. The placebo response is $\varphi(g_{\text{expectation}})$ evaluated under the expectation induced by being in a clinical trial.

Theorem 3.2 (Placebo = Gauge Zero). *[Full Proof] Under double-blind, placebo-controlled conditions, the **treatment effect estimate is gauge-invariant**:*

$$\hat{\theta}_{\text{true}} = \hat{\theta}_{\text{treatment}} - \hat{\theta}_{\text{placebo}}, \quad (8)$$

where $\hat{\theta}_{\text{placebo}}$ is the average outcome in the placebo arm. The placebo response cancels exactly because patients in both arms have the same expectation gatekeeper:

$$g_{\text{expectation}}^{(t)} = g_{\text{expectation}}^{(c)} = g_{\text{trial}}, \quad (9)$$

and thus the same gauge artifact $\varphi(g_{\text{trial}})$.

Proof. Step 1: Gauge equivalence of trial arms. Under double-blind conditions, patients in the treatment arm and the placebo arm have identical information about their treatment assignment. Both believe they might be receiving the active treatment, producing identical expectation gatekeepers:

$$g_{\text{expectation}}^{(t)} = g_{\text{expectation}}^{(c)} = g_{\text{trial}}. \quad (10)$$

Step 2: Observed outcomes. The observed outcome for a patient in the treatment arm is:

$$y_{\text{obs}}^{(t)} = y_{\text{true}}^{(t)} + \varphi(g_{\text{trial}}) + \varepsilon_t, \quad (11)$$

and for a patient in the placebo arm:

$$y_{\text{obs}}^{(c)} = y_{\text{true}}^{(c)} + \varphi(g_{\text{trial}}) + \varepsilon_c, \quad (12)$$

where $\varepsilon_t, \varepsilon_c$ are random noise.

Step 3: Gauge cancellation. The treatment effect estimate is:

$$\hat{\theta} = \bar{y}_{\text{obs}}^{(t)} - \bar{y}_{\text{obs}}^{(c)} = (\bar{y}_{\text{true}}^{(t)} - \bar{y}_{\text{true}}^{(c)}) + (\varphi(g_{\text{trial}}) - \varphi(g_{\text{trial}})) + (\bar{\varepsilon}_t - \bar{\varepsilon}_c) = \hat{\theta}_{\text{true}} + \text{noise}. \quad (13)$$

The gauge artifact $\varphi(g_{\text{trial}})$ cancels exactly. The placebo-controlled trial isolates the **gauge-invariant treatment effect**. $\square$

Corollary 3.3 (Placebo Effect Magnitude). *The magnitude of the placebo effect in a given trial is proportional to the **variance of the expectation gatekeeper** across patients:*

$$\|\hat{\theta}_{\text{placebo}}\| \propto \text{Var}(g_{\text{expectation}}). \quad (14)$$

This explains two empirical observations: (1) placebo effects are larger for subjective outcomes (pain, mood, self-reported function) where $\text{Var}(g_{\text{expectation}})$ is larger, and (2) placebo effects are smaller or absent for objective, hard outcomes (mortality, lab values) where $\text{Var}(g_{\text{expectation}}) \approx 0$ because the outcome is not expectation-sensitive.

3.2 The Placebo as the Audit Bench

The placebo arm serves a second function beyond gauge cancellation: it provides the **audit bench**—the reference distribution against which the treatment effect is measured.

Proposition 3.4 (Placebo Arm as Expert Bench). *In the SCX audit interpretation, the placebo arm is the **null-expert ensemble**: a set of expert reports generated under the same conditions as the treatment arm but without the active intervention. The Cercis score of the treatment effect is:*

$$S(\hat{\theta}) = -\log \mathbb{P}(\hat{\theta} \mid H_0 : \theta = 0), \quad (15)$$

where H_0 is estimated from the placebo arm data. Without a placebo arm, the audit bench is missing, and the **Cercis** score is undefined (or must be estimated from external historical data, introducing gauge artifacts).

4 Meta-Analysis as Cercis Computation

4.1 The Meta-Analytic Framework

Meta-analysis combines results from K independent trials to produce a single summary estimate [?, ?]. The two standard models are the fixed-effect model (all trials estimate the same underlying effect) and the random-effects model (trials estimate different effects drawn from a distribution).

Definition 4.1 (Meta-Analysis Ensemble). *A **meta-analysis ensemble** is a collection of K independent clinical trials $\{\mathcal{T}_1, \dots, \mathcal{T}_K\}$, each providing an effect estimate $\hat{\theta}_k$ with standard error σ_k . The ensemble has **audit multiplicity** $M_{meta} = \sum_{k=1}^K M_k$, where M_k is the audit multiplicity of trial k .*

Theorem 4.2 (Meta-Analysis Cercis = Weighted Sum of Trial Cercis Scores). *[Full Proof] The fixed-effect meta-analytic estimate is the **Cercis-weighted consensus** of the individual trial Cercis scores:*

$$\hat{\theta}_{FE} = \frac{\sum_k w_k \hat{\theta}_k}{\sum_k w_k}, \quad \text{where } w_k = \frac{1}{\sigma_k^2}. \quad (16)$$

The quality weight w_k is proportional to the **precision** of trial k (inverse variance), which is exactly the per-trial Cercis score:

$$S(\mathcal{T}_k) = -\log \mathbb{P}(\hat{\theta}_k \mid H_0) \propto \frac{\hat{\theta}_k^2}{2\sigma_k^2} = \frac{1}{2} \hat{\theta}_k^2 w_k. \quad (17)$$

The total meta-analytic Cercis score is:

$$S_{meta} = \sum_{k=1}^K S(\mathcal{T}_k) = \frac{1}{2} \sum_{k=1}^K \frac{\hat{\theta}_k^2}{\sigma_k^2} = \frac{1}{2} \hat{\theta}_{FE}^2 \sum_k w_k + \frac{1}{2} \sum_k w_k (\hat{\theta}_k - \hat{\theta}_{FE})^2, \quad (18)$$

decomposing into a consensus term (first summand) and a heterogeneity term (second summand).

Proof. Step 1: Likelihood under fixed-effect model. Under the fixed-effect model, all trials estimate a common effect θ . The log-likelihood across K independent trials is:

$$\log L(\theta) = -\frac{1}{2} \sum_{k=1}^K \frac{(\hat{\theta}_k - \theta)^2}{\sigma_k^2} + \text{const.} \quad (19)$$

Step 2: Maximum likelihood as Cercis optimization. Maximizing $\log L(\theta)$ is equivalent to minimizing the Cercis-weighted disagreement:

$$\hat{\theta}_{\text{FE}} = \arg \min_{\theta} \sum_{k=1}^K \frac{(\hat{\theta}_k - \theta)^2}{\sigma_k^2} = \frac{\sum_k \hat{\theta}_k / \sigma_k^2}{\sum_k 1 / \sigma_k^2}. \quad (20)$$

The variance-weighted average is the **Cercis consensus** of the trial-specific expert estimates.

Step 3: Decomposition of total Cercis. The per-trial log-likelihood contribution is exactly the Cercis quality score $Q(\mathcal{T}_k)$: it measures how surprising the trial result is under the null. Summing across trials and decomposing using $\hat{\theta}_k = \theta + (\hat{\theta}_k - \theta)$:

$$\sum_{k=1}^K \frac{\hat{\theta}_k^2}{\sigma_k^2} = \sum_{k=1}^K \frac{\theta^2}{\sigma_k^2} + \sum_{k=1}^K \frac{(\hat{\theta}_k - \theta)^2}{\sigma_k^2} + 2\theta \sum_{k=1}^K \frac{\hat{\theta}_k - \theta}{\sigma_k^2}. \quad (21)$$

At $\theta = \hat{\theta}_{\text{FE}}$, the cross-term vanishes, giving the decomposition in Eq. (??). $\square$

Corollary 4.3 (Forest Plot as Cercis Profile). *A **forest plot**—the standard visual display of meta-analysis—is a **Cercis profile** of the expert ensemble: each trial’s effect estimate $\hat{\theta}_k$ and confidence interval is an expert’s report with a quantifiable uncertainty (the variance σ_k^2). The overall diamond at the bottom is the Cercis-weighted consensus.*

4.2 Heterogeneity I^2 as Gauge Misalignment

Definition 4.4 (Heterogeneity Statistic I^2). *The I^2 statistic [?] measures the proportion of total variation across trials due to genuine heterogeneity rather than chance:*

$$I^2 = \frac{Q_{\text{total}} - (K - 1)}{Q_{\text{total}}} \times 100\%, \quad (22)$$

where $Q_{\text{total}} = \sum_{k=1}^K w_k (\hat{\theta}_k - \hat{\theta}_{\text{FE}})^2$ is Cochran’s Q statistic.

Proposition 4.5 (Heterogeneity $I^2 = \text{Gauge Misalignment}$). *I^2 is a measure of **gauge misalignment** across the trial expert ensemble. Specifically, I^2 estimates the proportion of variance in the Cercis score that is due to gauge-dependent differences across trials (study-level factors) rather than gauge-invariant sampling error:*

$$I^2 = \frac{\tau^2}{\tau^2 + s^2} \times 100\%, \quad (23)$$

where τ^2 is the between-trial variance (estimated via DerSimonian–Laird or REML) and s^2 is a typical within-trial variance. In SCX terms:

$$I^2 = \frac{\|\text{gauge misalignment}\|^2}{\|\text{gauge misalignment}\|^2 + \|\text{audit resolution}\|^2} \times 100\%. \quad (24)$$

An I^2 near 0% indicates all trials are gauge-equivalent; an I^2 near 100% indicates each trial operates under a different gauge.

Proof Sketch. Under the random-effects model, the observed treatment effect in trial k is $\hat{\theta}_k = \theta + u_k + \varepsilon_k$, where $u_k \sim \mathcal{N}(0, \tau^2)$ is the gauge deviation (between-trial heterogeneity) and $\varepsilon_k \sim \mathcal{N}(0, \sigma_k^2)$ is the sampling noise. Gauge alignment corresponds to $u_k = 0$ for all k —all trials measure the same invariant effect. Gauge misalignment $u_k \neq 0$ arises from differences in patient populations, outcome definitions, concomitant care, and other study-level factors that change the “gauge”—the measurement context—across trials.

The I^2 statistic estimates the proportion of the total variance $\text{Var}(\hat{\theta}_k) = \tau^2 + \sigma_k^2$ that is gauge-driven (τ^2). In the limit $\tau^2 \rightarrow 0$, all trials are perfectly gauge-aligned and $I^2 \rightarrow 0$. In the limit $\tau^2 \rightarrow \infty$, trials are completely gauge-independent and $I^2 \rightarrow 1$. $\square$

Remark 4.6 (Subgroup Analysis as Gauge Restoration). Subgroup analysis in meta-analysis—testing whether treatment effects differ by patient age, sex, disease severity, etc.—is an attempt to **restore gauge invariance** by conditioning on the gauge-relevant variables. If a subgroup definition absorbs the source of heterogeneity ($\tau_{\text{within}}^2 \ll \tau_{\text{between}}^2$), the gauge alignment is restored and I^2 drops.

4.3 Random-Effects as Annealing Schedule

Proposition 4.7 (Random-Effects = Cercis Annealing). *The random-effects meta-analysis model corresponds to the Cercis score with positive annealing schedule $\eta(t) > 0$:*

$$\hat{\theta}_{RE} = \frac{\sum_k w_k^* \hat{\theta}_k}{\sum_k w_k^*}, \quad w_k^* = \frac{1}{\sigma_k^2 + \hat{\tau}^2}, \quad (25)$$

where $\hat{\tau}^2$ is the estimated heterogeneity variance. The Cercis annealing parameter η is proportional to the total heterogeneity τ^2 : larger heterogeneity means larger η , which reduces the weight of individual trials (experts) and spreads the consensus more evenly—the analog of a higher “temperature” in the Cercis annealing schedule.

5 Diagnostic Disagreement and the Cercis Signal

5.1 Radiologist Disagreement

Diagnostic disagreement among expert radiologists is well-documented and substantial. In mammography screening, the inter-rater agreement kappa between two radiologists interpreting the same mammogram is typically $\kappa = 0.4$ – 0.6 [?, ?]. For breast biopsy interpretation, agreement can be as low as $\kappa = 0.3$ [?].

Definition 5.1 (Diagnostic Disagreement as Cercis Signal). *Let $d_1, d_2 \in \{0, 1\}$ be the binary diagnostic decisions (positive or negative) of two independent experts evaluating the same patient. The **diagnostic Cercis score** is:*

$$S_{diag} = -\log \mathbb{P}(d_1 = d_2 \mid H_0), \quad (26)$$

where H_0 is the null hypothesis that both experts are random guessers. High disagreement (low kappa) produces a high Cercis score because the probability of two independent experts agreeing by chance is low when the base rate is extreme.

Theorem 5.2 (Second Opinion as $M + 1$ Audit). *[Full Proof] A second opinion from an independent expert is an $M + 1$ **audit extension**: if the first expert (gatekeeper g_1) provides a diagnosis with bias $\Delta_1 = \|g_1\|$, the second expert (gatekeeper g_2) adds an*

independent audit dimension. The joint false-positive rate after two independent positive diagnoses satisfies:

$$FPR_{1\&2} = FPR_1 \cdot FPR_2 + \text{Cov}(d_1, d_2 \mid H_0), \quad (27)$$

where $\text{Cov}(d_1, d_2 \mid H_0)$ is the covariance in false-positive tendencies. Under conditional independence given the true state:

$$FPR_{1\&2} = FPR_1 \cdot FPR_2. \quad (28)$$

For typical $FPR_1 = FPR_2 = 0.1$, the joint FPR is 0.01—a tenfold reduction. The second opinion thus **exponentially reduces false positives** when experts are conditionally independent.

Proof. Step 1: Cercis score of disagreement. Under the null hypothesis H_0 that the patient is disease-free, the probability that two independent experts both give positive diagnoses is:

$$\mathbb{P}(d_1 = 1, d_2 = 1 \mid H_0) = \mathbb{P}(d_1 = 1 \mid H_0) \cdot \mathbb{P}(d_2 = 1 \mid H_0) + \text{Cov}(d_1, d_2 \mid H_0). \quad (29)$$

If the experts are conditionally independent given the true state (the standard assumption in diagnostic test evaluation), the covariance is zero.

Step 2: Audit depth increase. The first expert provides an audit of depth $M = 1$: a single expert’s opinion. The second expert increases the audit depth to $M = 2$. In the SCX framework, the Cercis score for M independent experts is:

$$S(d_1, \dots, d_M) = -\log \mathbb{P}(d_1, \dots, d_M \mid H_0) = \sum_{i=1}^M -\log \mathbb{P}(d_i \mid H_0). \quad (30)$$

The confidence in a consensus diagnosis grows exponentially with M .

Step 3: Gatekeeper reduction. The first expert has gatekeeper g_1 (their individual bias). The consensus of two experts has a reduced effective gatekeeper:

$$\|g_{\text{consensus}}\| \leq \frac{\|g_1\| \cdot \|g_2\|}{\sqrt{\|g_1\|^2 + \|g_2\|^2}}, \quad (31)$$

which is always less than either individual gatekeeper norm. When $g_1 = g_2 \neq 0$, $\|g_{\text{consensus}}\| = \|g\|/\sqrt{2}$, representing a $\sqrt{M}$ reduction in bias for M independent experts. $\square$

Corollary 5.3 (Diagnostic Consensus Committees). *A diagnostic consensus committee of M independent experts achieves a Cercis score that scales as $S_{\text{cons}} \propto M \cdot S_{\text{single}}$. This explains the standard practice of using consensus panels (e.g., tumor boards) for complex diagnoses: they multiply the audit depth.*

5.2 Double Reading in Screening

In many cancer screening programs (e.g., mammography in the UK NHS Breast Screening Programme), double reading—two radiologists independently interpret each screening mammogram—is standard practice.

Proposition 5.4 (Double Reading as $M = 2$ Audit). *Double reading is a canonical $M = 2$ audit. The **Cercis** score improves by approximately 2–10% in sensitivity and specificity compared to single reading, depending on the radiologists’ experience and the screening context [?]. This corresponds to a reduction in the effective gatekeeper norm by a factor of $1/\sqrt{2}$:*

$$NNT_{\text{second reader}} = \frac{1}{FPR_{\text{single}} - FPR_{\text{double}}}, \quad (32)$$

the number needed to screen to prevent one false-positive recall via double reading. Empirical estimates place $NNT_{\text{second reader}} \approx 100\text{--}500$ in mammography.

6 The Evidence Hierarchy as Audit Depth

The Oxford Centre for Evidence-Based Medicine (OCEBM) hierarchy ranks study designs by their susceptibility to bias [?]. We show that this hierarchy corresponds to a mathematical **audit depth function**.

Definition 6.1 (Evidence Hierarchy as Audit Depth Function). *The **audit depth function** $\text{AuditDepth} : \text{StudyTypes} \rightarrow \mathbb{R}^+$ maps each study design to its effective audit multiplicity:*

$$\text{AuditDepth}(\text{study}) = M_{\text{eff}} = \frac{1}{\|g_{\text{study}}\|} \geq 1, \quad (33)$$

where $\|g_{\text{study}}\|$ is the gatekeeper norm of the study configuration. The hierarchy from highest to lowest is:

<i>Systematic review / meta-analysis</i>	:	$M_{\text{eff}} \gg 1, \ g\ \approx \tau_{\text{meta}}$	
<i>RCT (double-blind)</i>	:	$M_{\text{eff}} \approx n_{\text{eff}}, \ g\ \approx 1/\sqrt{n_{\text{eff}}}$	
<i>Cohort study</i>	:	$M_{\text{eff}} < n_{\text{eff}}, \ g\ = \ \Gamma\ _{\text{confound}}$	(34)
<i>Case-control study</i>	:	$M_{\text{eff}} \ll n_{\text{eff}}, \ g\ = \ \Gamma\ _{\text{recall}}$	
<i>Case series / case report</i>	:	$M_{\text{eff}} \approx 1, \ g\ = \ g_{\text{author}}\ $	
<i>Expert opinion</i>	:	$M_{\text{eff}} = 1, \ g\ = \infty$ (self-certifying).	

Theorem 6.2 (EBM Hierarchy = Audit Depth Monotonicity). *[Full Proof]The OCEBM evidence hierarchy is strictly monotonic in audit depth:*

$$\text{Level } 1 \succ \text{Level } 2 \succ \dots \succ \text{Level } 5 \iff M_{\text{eff}}^{(1)} > M_{\text{eff}}^{(2)} > \dots > M_{\text{eff}}^{(5)}, \quad (35)$$

*where $\succ$ denotes “superior evidence” and $M_{\text{eff}}^{(i)}$ is the effective audit multiplicity at evidence level i . The **Cercis** score of a study at level i is proportional to $M_{\text{eff}}^{(i)}$:*

$$S(\text{study at level } i) \propto \frac{\theta^2}{2\sigma^2} \cdot M_{\text{eff}}^{(i)}, \quad (36)$$

meaning that higher-level studies produce exponentially more evidence per unit effect size.

Proof. Step 1: Gatekeeper norms. We compute $\|g\|$ for each study type:

leftmargin=* Systematic reviews: $\|g\| = \tau_{\text{meta}}$ (between-study heterogeneity). For well-conducted meta-analyses, τ_{meta} is estimated via random-effects, giving $\|g\| \ll 1$ when heterogeneity is low.

- leftmargin=* Double-blind RCTs: $\|g\| \approx 1/\sqrt{n}$ by the central limit theorem. The gatekeeper norm decreases as $1/\sqrt{n}$ with increasing sample size.
- leftmargin=* Cohort studies: $\|g\| = \|\Gamma\|$, the norm of the confounding vector. This is typically 0.1–0.5 in well-conducted studies but does not decrease with n —confounding is a structural bias, not a sampling bias.
- leftmargin=* Case-control studies: $\|g\| = \|\Gamma\|_{\text{recall}}$ includes recall bias and selection bias, typically larger than confounding.
- leftmargin=* Case series: $\|g\| = \|g_{\text{author}}\|$, the author’s personal bias, unconstrained by any systematic comparison.
- leftmargin=* Expert opinion: $\|g\| \rightarrow \infty$ because the expert self-certifies.

Step 2: Monotonicity. The ordering $M_{\text{eff}}^{(i)} = 1/\|g\|^{(i)}$ gives a strict monotonic decrease from systematic reviews ($\|g\|$ smallest) to expert opinion ($\|g\| \rightarrow \infty$). This matches the OCEBM hierarchy exactly: the level ordering corresponds to increasing gatekeeper norm, equivalently decreasing audit depth.

Step 3: Cercis scaling. The Cercis score of a study type scales as $S \propto M_{\text{eff}}$ because the effective sample size—the number of independent expert reports—is M_{eff} . For a fixed true effect θ , the log-likelihood ratio evidence is:

$$\log \Lambda = \sum_{i=1}^{M_{\text{eff}}} \log \frac{\mathbb{P}(y_i | \theta)}{\mathbb{P}(y_i | 0)} \asymp M_{\text{eff}} \cdot \text{KL}(\theta \parallel 0), \quad (37)$$

where KL is the Kullback–Leibler divergence between the treatment and null distributions for a single observation. $\square$

Corollary 6.3 (GRADE as Audit Certainty). *The GRADE system (Grading of Recommendations, Assessment, Development and Evaluations) [?] rates the certainty of evidence as high, moderate, low, or very low. In SCX terms, GRADE certainty is:*

$$\text{Certainty} \propto \frac{1}{\|g\|} = M_{\text{eff}}. \quad (38)$$

*Factors that decrease certainty (risk of bias, inconsistency, indirectness, imprecision, publication bias) are all **gauge-inflation mechanisms** that increase $\|g\|$.*

7 Eminence-Based Medicine: The $M = 1$ Catastrophe

Eminence-based medicine—clinical decisions based on the opinion of a senior expert rather than on systematic evidence—is a standing target of critique within EBM [?]. In SCX terms, it is the $M = 1$ audit catastrophe.

Definition 7.1 (Eminence-Based Medicine). ***Eminence-based medicine** is a clinical decision-making protocol with audit multiplicity $M = 1$: a single expert (the “eminence”) provides a diagnosis, treatment recommendation, or prognosis without external audit. The Cercis score is:*

$$S_{\text{eminence}} = -\log \mathbb{P}(d_{\text{eminence}} | H_0), \quad (39)$$

but critically, H_0 is self-specified by the same expert: the eminence defines both the hypothesis and the evidence, making the audit circular.

Theorem 7.2 (The $M = 1$ Catastrophe). *[Full Proof] A single-expert audit ($M = 1$) with gatekeeper g has **unbounded false-positive rate** in the worst case:*

$$\sup_{\|g\| \rightarrow \infty} FPR_{M=1} = 1. \quad (40)$$

Moreover, the expected *Cercis* score of a single self-auditing expert is identically zero under any non-degenerate prior:

$$\mathbb{E}[S_{\text{eminence}}] = 0. \quad (41)$$

This means eminence-based medicine provides zero expected evidence for the correctness of its recommendations.

Proof. Step 1: Self-certification as gauge divergence. A single expert who both makes a claim and audits it has gatekeeper norm $\|g\| \rightarrow \infty$ because there is no external constraint on their internal state. The claim d_{eminence} is always consistent with the self-specified null because the expert can adjust H_0 to fit any outcome.

Step 2: False-positive bound. For any $p \in (0, 1)$, the eminence can claim $d = 1$ with “ p -value” less than p by constructing a null that makes the observation $d = 1$ maximally improbable. In formal terms, the supremum of the false-positive rate over all possible self-specifications of H_0 is 1.

Step 3: Zero expected evidence. The expected *Cercis* score is:

$$\mathbb{E}[S_{\text{eminence}}] = \mathbb{E}[-\log \mathbb{P}(d_{\text{eminence}} \mid H_0)]. \quad (42)$$

But because the expert defines H_0 after observing the data (or equivalently chooses both simultaneously), $\mathbb{P}(d_{\text{eminence}} \mid H_0)$ can be made arbitrarily close to 1 for any outcome, giving $S \approx 0$ in expectation. More formally, the data-processing inequality ensures that no information can be extracted from a self-auditing channel. $\square$

Corollary 7.3 (Why EBM Replaced Eminence). *The replacement of eminence-based medicine by evidence-based medicine in the late 20th century was not a sociological shift but an **epistemological necessity**: any decision-making system with audit multiplicity $M = 1$ produces zero expected evidence, and any system with $M > 1$ produces strictly positive expected evidence. The history of medicine from 1950–2000 is the history of increasing M .*

7.1 The Transition from $M = 1$ to $M > 1$

Historically, medicine operated largely at $M = 1$: the physician’s opinion was the standard of care. The clinical trial revolution increased M dramatically:

Proposition 7.4 (Historical M Growth). *The effective audit multiplicity in clinical medicine has grown from $M \approx 1$ in 1900 to $M \gg 10^4$ in modern Phase III RCTs. Key milestones:*

$$\begin{aligned} 1900 & : M \approx 1 \text{ (physician opinion)} \\ 1948 & : M = 107 \text{ (Streptomycin trial, first proper RCT)} \\ 1960s & : M \sim 10^2\text{--}10^3 \text{ (multi-center RCTs)} \\ 2000s & : M \sim 10^3\text{--}10^4 \text{ (megatrials, e.g., ISIS, HPS)} \\ 2020s & : M \sim 10^4\text{--}10^5 \text{ (platform trials, adaptive trials).} \end{aligned} \quad (43)$$

Each order-of-magnitude increase in M corresponds to an approximately $\sqrt{M}$ -fold reduction in the standard error and an exponential increase in the *Cercis* score.

8 Reproducibility Crisis as Audit Failure

The reproducibility crisis—the observation that many published research findings cannot be replicated [?, ?]—is an **audit failure** in the SCX framework.

8.1 p -Hacking as Gauge Manipulation

Definition 8.1 (p -hacking). ***p -hacking** (also called *significance chasing*, *data dredging*, or *fishing*) is the practice of selectively reporting analyses, outcomes, or subgroups that achieve statistical significance ($p < 0.05$) while suppressing non-significant results [?].*

Theorem 8.2 (p -Hacking = Gauge Manipulation). *[Full Proof] p -hacking is mathematically equivalent to a ***gauge transformation*** on the audit configuration. Specifically, let $\mathcal{T}$ be a trial with gatekeeper g , and let $\mathcal{T}'$ be the p -hacked version (reported outcomes selectively chosen). Then:*

$$g' = g + \delta_{\text{hack}}, \quad (44)$$

where δ_{hack} is the ***gauge shift*** from p -hacking. The false-positive rate under p -hacking inflates by exactly:

$$\text{FPR}_{\text{hacked}} = \alpha \cdot e^{D_{\text{hack}}}, \quad (45)$$

where α is the nominal significance level and D_{hack} is the effective number of independent “degrees of hacking” (analyses tried, outcomes examined, subgroups tested):

$$D_{\text{hack}} = \sum_{a \in \mathcal{A}} w_a \cdot \mathbb{1}[\text{analysis } a \text{ attempted but not significant}], \quad (46)$$

with $\mathcal{A}$ the set of all analyses considered and w_a the effective number of independent comparisons per analysis.

Proof. Step 1: Gauge structure of the null. Under the null hypothesis H_0 , the test statistic T follows a known distribution. The decision rule “reject H_0 if $|T| > t_\alpha$ ” defines a gauge: it is a specific mapping from data to decision. p -hacking changes this mapping by allowing the analyst to choose which test statistic T to report.

Step 2: Multiple comparisons as gauge transformations. Each attempted analysis $a \in \mathcal{A}$ defines a random variable T_a under H_0 . The p -hacked analyst selects the minimal p -value:

$$p_{\min} = \min_{a \in \mathcal{A}} \mathbb{P}(|T_a| \geq |T_a^{\text{obs}}| \mid H_0). \quad (47)$$

This is a gauge transformation because it changes the effective null distribution: under the reporting gauge defined by p -hacking, the distribution of the reported p -value is no longer Uniform[0, 1] but:

$$\mathbb{P}(p_{\text{reported}} < \alpha) = 1 - (1 - \alpha)^{D_{\text{hack}}} \approx \alpha D_{\text{hack}} \quad \text{for small } \alpha. \quad (48)$$

Step 3: Gauge shift magnitude. The gauge shift δ_{hack} has norm proportional to the excess false-positive rate:

$$\|\delta_{\text{hack}}\| \propto \log \left(\frac{\text{FPR}_{\text{hacked}}}{\alpha} \right) = \log D_{\text{hack}}. \quad (49)$$

This shows that even moderate p -hacking ($D_{\text{hack}} = 10$, corresponding to about 10 analyses) produces an inflation of $\text{FPR} \approx 10\alpha = 0.5$ at $\alpha = 0.05$ —the false positive rate approaches chance. $\square$

Corollary 8.3 (*p*-Curve as Gauge Diagnostic). *Simonsohn’s p-curve analysis [?]*—examining the distribution of significant *p*-values in a published literature—is a ***gauge diagnostic***. A *p*-curve that is right-skewed indicates no *p*-hacking (gauge-stable), while a left-skewed or flat *p*-curve indicates *p*-hacking (gauge-shifted). The *p*-curve is an empirical estimate of D_{hack} .

8.2 Publication Bias as Censored Audit

Definition 8.4 (Publication Bias as Audit Censoring). **Publication bias** is the preferential publication of studies with statistically significant results [?]. In SCX terms, it is **audit censoring**: the observed expert ensemble (published trials) is a systematically biased subset of the full expert ensemble (all conducted trials). The censoring mechanism is:

$$\mathbb{P}(\text{published} \mid \mathcal{T}_k) \propto \mathbb{P}(p_k < 0.05 \mid \mathcal{T}_k), \quad (50)$$

meaning the probability of observing trial *k* in the meta-analytic audit is proportional to its significance.

Proposition 8.5 (Funnel Plot as Audit Completeness Diagnostic). A funnel plot—effect size plotted against standard error—is an **audit completeness diagnostic**. In the absence of publication bias, the funnel should be symmetric: small trials (large SE) scatter widely around the pooled estimate, while large trials cluster near it. Asymmetric funnels indicate audit censoring. The Egger test [?] for funnel asymmetry is a test for whether the censored audit differs systematically from the complete audit:

$$\text{Egger intercept} \propto \|\text{censoring bias}\|. \quad (51)$$

8.3 The Replication Audit

Theorem 8.6 (Replication as Independent Audit). **[Full Proof]** A replication study is an **independent re-audit** of the same scientific claim. If the original study has effect estimate $\hat{\theta}_1$ with variance σ_1^2 , and the replication has $\hat{\theta}_2$ with variance σ_2^2 , then the replication Cercis score is:

$$S_{\text{replication}} = -\log \mathbb{P}(|\hat{\theta}_2 - \hat{\theta}_1| \geq |\hat{\theta}_2^{obs} - \hat{\theta}_1| \mid H_0 : \theta_1 = \theta_2). \quad (52)$$

Under the null that the original finding is real, the combined evidence after *R* replications is:

$$S_{\text{combined}} = \sum_{r=1}^R S_{\text{replication}}^{(r)}. \quad (53)$$

The reproducibility rate across a field—the proportion of replication studies that achieve $p < 0.05$ in the same direction—measures the **audit reliability** of that field’s published literature. An estimated reproducibility rate of 36% in psychology [?] indicates severe audit failure: the modal published finding is likely a false positive.

9 Cercis-Based Trial Design

The SCX framework provides a principled approach to designing clinical trials: choose the audit multiplicity *M* to maximize the expected Cercis score for a given effect size.

9.1 Optimal Audit Multiplicity

Definition 9.1 (Optimal Audit Multiplicity). *The **optimal audit multiplicity** M^* for detecting a true effect of size θ with a null model of variance σ_0^2 is the minimum M that achieves a target Cercis score S^* :*

$$M^*(\theta, S^*) = \min\{M \in \mathbb{N} : \mathbb{E}[S_M(\hat{\theta})] \geq S^*\}, \quad (54)$$

where $S_M(\hat{\theta})$ is the Cercis score achieved by an M -expert audit for effect $\hat{\theta}$.

Proposition 9.2 (Sample Size as Audit Multiplicity). *The conventional sample size calculation for a clinical trial is a special case of the optimal audit multiplicity problem. For a continuous outcome with effect size $d = \theta/\sigma$, the required sample size per arm is:*

$$n = \frac{2(z_{\alpha/2} + z_{\beta})^2}{d^2}, \quad (55)$$

where $z_{\alpha/2}$ and z_{β} are the standard normal quantiles for the Type I and Type II error rates. In Cercis terms:

$$M^* = 2n = \frac{4(z_{\alpha/2} + z_{\beta})^2}{d^2}, \quad (56)$$

and the expected Cercis score at the true effect size is:

$$\mathbb{E}[S(\hat{\theta})] = \frac{nd^2}{2} = \frac{M^*d^2}{4}. \quad (57)$$

Thus M^* scales as $1/d^2$ for small effects: detecting small effects requires vastly larger audit multiplicities.

9.2 Sequential Monitoring as Sequential Audit

Modern clinical trials often employ sequential monitoring—interim analyses at pre-specified information fractions [?]. This corresponds to **sequential audit** in the SCX framework.

Definition 9.3 (Group Sequential Audit). *A **group sequential audit** of a clinical trial with L analyses at information fractions $\pi_1 < \pi_2 < \dots < \pi_L = 1$ produces a sequence of Cercis scores $S_1, \dots, S_L$, where:*

$$S_{\ell} = -\log \mathbb{P}(\hat{\theta}_{\ell} \geq \hat{\theta}_{\ell} \mid H_0 : \theta = 0), \quad (58)$$

and $\hat{\theta}_{\ell}$ is the cumulative effect estimate at analysis ℓ . The stopping boundary—the threshold for early stopping—defines an **audit stopping rule**:

$$\tau_{stop} = \min\{\ell \leq L : S_{\ell} \geq c_{\ell}(\alpha)\}, \quad (59)$$

where $c_{\ell}(\alpha)$ is the Cercis threshold at analysis ℓ , chosen to preserve the overall Type I error rate α .

Proposition 9.4 (O’Brien-Fleming as Constant-Cercis Boundary). *The O’Brien-Fleming stopping boundary [?]
—the most commonly used sequential monitoring boundary—is a **constant-Cercis** boundary:*

$$z_{\ell}(\alpha) = \frac{z_{\alpha/2}}{\sqrt{\pi_{\ell}}}, \quad c_{\ell}(\alpha) = \frac{z_{\alpha/2}^2}{2\pi_{\ell}}. \quad (60)$$

At each analysis, the required Cercis score per information fraction is constant: $\pi_{\ell} \cdot c_{\ell}(\alpha) = z_{\alpha/2}^2/2$. This makes the O’Brien-Fleming boundary the unique sequential audit that spreads the audit effort evenly across the trial duration.

10 Discussion

10.1 Summary of Results

We have established nine formal correspondences between clinical trials and the SCX audit framework, summarized in Table ??.

Table 1: Summary of clinical trial–SCX audit correspondences.

#	Clinical Trial Concept	SCX Audit Analog
1	Phase III RCT	$M > 1$ expert audit
2	Randomization	Gauge fixing
3	Blinding	Gatekeeper zeroing
4	Placebo control	Gauge zero (bench)
5	Meta-analysis	Cercis computation
6	I^2 heterogeneity	Gauge misalignment measure
7	Diagnostic disagreement	High-Cercis signal
8	EBM hierarchy	Audit depth function
9	Eminence-based medicine	$M = 1$ catastrophe
10	p -hacking	Gauge manipulation
11	Publication bias	Censored audit
12	Sequential monitoring	Sequential audit

10.2 What This Reframing Gives Us

Unified language. The SCX framework provides a single mathematical language for discussing bias, evidence, and consensus across all study designs, from case reports to network meta-analyses.

Quantitative audit depth. The audit depth function gives a precise, quantitative measure of evidence quality that goes beyond ordinal hierarchies. One can compute the expected Cercis score per patient in a trial design and compare it across designs.

Cercis-optimized design. Trial designers can choose the sample size, number of centers, investigators per center, and outcome granularity to maximize the expected Cercis score per unit cost—an optimization problem that conventional sample size calculations do not address.

Detection of p -hacking. The gauge manipulation theory provides a quantitative prediction for the excess false-positive rate under hacking, enabling prospective correction.

New approach to placebo. Understanding the placebo response as a gauge artifact suggests that placebo-controlled trials could be replaced (in some contexts) by gauge-adjusted analyses, reducing the number of patients exposed to placebo.

10.3 What This Paper Does *Not* Claim

- leftmargin=* **We do not claim** that clinical trials are literally identical to the SCX audit framework in all respects. The mapping is a structural isomorphism at the level of bias, evidence aggregation, and consensus formation, not a claim that trial participants are literally “experts” in the SCX sense in all dimensions.
- leftmargin=* **We do not claim** that the Cercis score should replace conventional p -values or confidence intervals in clinical trial reporting. The Cercis score provides a complementary information-theoretic perspective that illuminates the underlying audit structure.
- leftmargin=* **We do not claim** that all medicine can be reduced to audit theory. Clinical judgment, patient values, and qualitative considerations are irreducible components of medical decision-making.

10.4 Open Problems

- leftmargin=*, label=(1) **Audit-optimal trial design.** What is the optimal allocation of a fixed budget B across sample size n , number of centers C , and number of investigators per center I to maximize the expected Cercis score? This is a convex optimization problem in the SCX framework.
- leftmargin=*, label=(2) **Gauge-equivalent endpoints.** Are there pairs of clinical endpoints that are gauge-equivalent—i.e., that produce the same Cercis score for a given treatment effect? If so, trials using different but gauge-equivalent endpoints could be directly meta-analyzed without standardization.
- leftmargin=*, label=(3) **Blinding effectiveness as gatekeeper estimation.** Can the effectiveness of blinding in a clinical trial be quantified as an estimated gatekeeper norm $\|\hat{g}\|$? This would provide a quantitative measure of trial quality that supplements the Jadad score or the Cochrane risk-of-bias tool.
- leftmargin=*, label=(4) **Precision medicine as individual audit.** Does precision medicine—tailoring treatments to individual patient characteristics—correspond to an $M \rightarrow 1$ transition? That is, does the audit multiplicity shrink from $M_{\text{population}}$ to $M_{\text{individual}} = 1$ when treating the individual, reinstituting the $M = 1$ catastrophe at the individual level?
- leftmargin=*, label=(5) **Bayesian Cercis for trials.** How does Bayesian trial design (using priors to incorporate historical evidence) modify the Cercis score computation? The prior corresponds to a pre-audit Cercis score from previous trials.
- leftmargin=*, label=(6) **Diagnostic test evaluation as audit.** The evaluation of a diagnostic test measures its sensitivity and specificity relative to a gold standard. In SCX terms, the gold standard is an audit authority with gatekeeper $g = 0$, and the test under evaluation

has $g > 0$. The AUC (area under the ROC curve) is a measure of the test’s audit quality.

10.5 Philosophical Implications

The identification of clinical trials with $M > 1$ audits suggests that evidence-based medicine is not merely a methodological preference but an epistemological necessity: any belief-forming system that relies on a single expert ($M = 1$) produces zero expected evidence. The historical transition from eminence-based medicine to evidence-based medicine is the story of increasing M —from the solitary physician at the bedside to the multi-center, multi-national, multi-thousand-patient mega-trial.

If this perspective is correct, then the core insight of evidence-based medicine—that aggregate evidence from many independent sources is more reliable than the opinion of any single authority—is a special case of the general principle that multi-expert audit is exponentially more informative than self-certification. The SCX framework does not replace EBM; it provides the mathematical foundation that EBM has been missing.

Appendix A: Glossary of Notation

Symbol	Meaning
$\mathcal{T} = (\mathcal{P}, \mathcal{I}, \mathcal{D}, \mathcal{E})$	Clinical trial audit configuration
M	Audit multiplicity (number of independent expert auditors)
$\hat{\theta}_k$	Treatment effect estimate in trial k
σ_k^2	Variance of $\hat{\theta}_k$
$w_k = 1/\sigma_k^2$	Inverse-variance weight (Cercis weight)
I^2	Heterogeneity statistic (gauge misalignment)
τ^2	Between-trial variance
$\hat{\theta}_{\text{FE}}$	Fixed-effect meta-analytic estimate
$\hat{\theta}_{\text{RE}}$	Random-effects meta-analytic estimate
g	Gatekeeper parameter (concealable bias)
$\ g\ $	Gatekeeper norm
$\varphi(g)$	Gauge transformation (placebo response)
δ_{hack}	Gauge shift from p -hacking
D_{hack}	Effective degrees of hacking
$S(s) = Q(s) + \eta(t)N(s)$	Cercis score
$Q(s)$	Quality (agreement) component
$N(s)$	Novelty (distinguishability) component
$\eta(t)$	Annealing schedule
<code>AuditDepth(study)</code>	Audit depth function
FPR	False-positive rate
κ	Cohen’s kappa (inter-rater agreement)

Appendix B: Numerical Verification

The Python script `verify_medicine_audit.py` accompanying this paper provides numerical verification of the key quantitative claims:

- leftmargan=*, label=(1) **Cercis meta-analysis computation:** The equivalence between the fixed-effect meta-analytic estimate and the Cercis-weighted consensus is verified by simulating K trials and comparing the weighted estimate to the Cercis score maximum.
- leftmbrgbn=*, lbbel=(2) **I^2 as gauge misalignment:** The relationship between I^2 and the between-trial variance τ^2 is verified by simulating trials with controlled heterogeneity and measuring I^2 .
- leftmrgcn=*, lcbel=(3) **p -hacking detection via Cercis:** The inflation of the false-positive rate under p -hacking is verified by simulating multiple analyses and measuring the excess false positives.
- leftmdrgdn=*, ldbel=(4) **Diagnostic disagreement modeling:** The exponential reduction in FPR from second opinions is verified by simulating conditionally independent expert diagnoses.
- leftmergen=*, lebel=(5) **Optimal M calculation:** The scaling of optimal sample size with effect size is verified by computing M^* across a range of effect sizes.
- leftmfrgfn=*, lfbel=(6) **Evidence hierarchy audit depth:** The monotonic relationship between study type and audit multiplicity is verified by computing M_{eff} for each study design.
- leftmgrggn=*, lgbel=(7) **Sequential trial monitoring:** The constant-Cercis property of the O'Brien-Fleming boundary is verified by computing the cumulative Cercis score at each interim analysis.

All numerical claims in the paper can be reproduced by running:

```
python verify_medicine_audit.py
```

with `numpy` and `scipy` installed. Expected runtime on a modern laptop: ~ 15 seconds.

Appendix C: Proof of $\mathbb{E}[S_{\text{eminence}}] = 0$

Theorem 10.1 (Zero Expected Evidence for $M = 1$ Self-Audit). *[Full Proof] For any single-expert audit ($M = 1$) where the expert defines both the hypothesis H_0 and the evidence (data x), the expected Cercis score is zero:*

$$\mathbb{E}_{x|H_0}[S(x, H_0(x))] = 0, \quad (61)$$

where $H_0(x)$ indicates that the null hypothesis may depend on the observed data.

Proof. Step 1: Self-audit as data-dependent hypothesis. In a self-audit, the expert observes data x , then formulates a null hypothesis $H_0(x)$ (or equivalently, reports a p -value $p(x)$ that depends on x). The reported Cercis score is:

$$S_{\text{reported}} = -\log p(x). \quad (62)$$

Step 2: The expert can always maximize p . Because the expert controls H_0 , they can choose H_0 such that x appears typical under H_0 . The supremum over H_0 of $p(x | H_0)$ is at least 1 (by choosing H_0 as the empirical distribution of x). Thus:

$$\sup_{H_0} \mathbb{P}(x | H_0) = 1 \implies \inf_{H_0} S(x, H_0) = 0. \quad (63)$$

Step 3: Expectation. The expected Cercis score is:

$$\mathbb{E}[S] = \int p(x)(-\log p_{\text{reported}}(x)) dx, \quad (64)$$

where $p_{\text{reported}}(x)$ is the expert’s reported p -value. Since the expert can always report $p_{\text{reported}}(x) = 1$, $\mathbb{E}[S] \geq 0$. Conversely, by the information inequality, the expected log-likelihood ratio is $\mathbb{E}[\log p/q] \leq 0$ when p is the true distribution and q is the reported one. For a self-audit, $p = q$ (the expert uses the data distribution as the null), giving $\mathbb{E}[S] = \mathbb{E}[-\log(p(x)/p(x))] = 0$.

Step 4: Strictly zero. The expected Cercis score is zero, and any positive reported Cercis score in a single self-audit is the result of the expert choosing not to use the optimal H_0 —i.e., the expert chooses to appear informative by selecting a H_0 that makes x unlikely, which is exactly the definition of p -hacking. $\square$

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